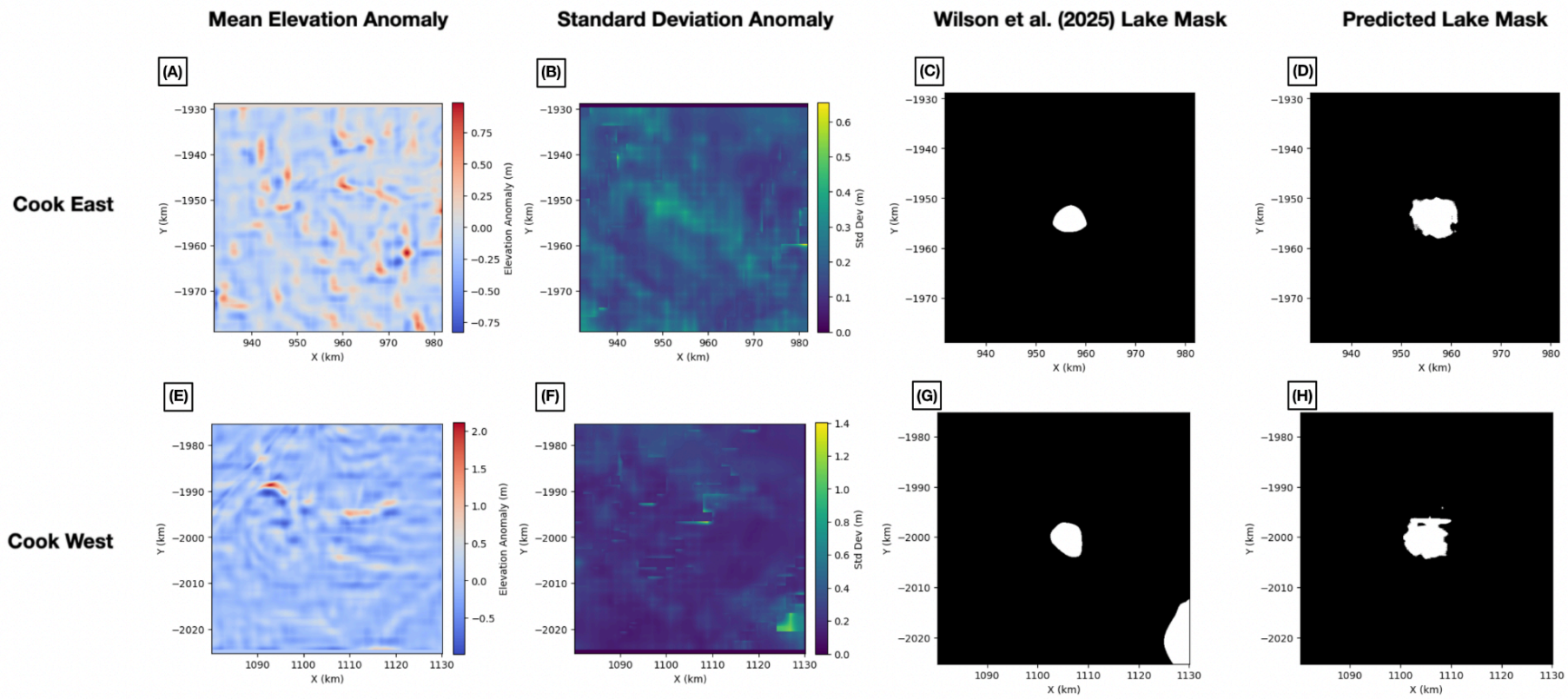
Figure 1. Model input features and lake mask predictions for Cook East (top row; A – D), and Cook West (bottom row; E – H). All panels show spatial coordinates (km) in the polar stereographic projection. Panels (A, E) show the mean elevation anomaly drive from the CryoSat-2 elevation time series, and Panels (B, F) show the standard deviation of elevation anomalies. Panels (C,G) show the "observed" subglacial lake masks derived from the Wilson et al. (2025) validation set. Panels (D,H) show the corresponding lake masks predicted by the CNN U-Net Round 1 Model, where the elevation anomaly statistics were used as input features. Regions with surface elevation anomalies correlated with observed subglacial lakes, where the model predicted the approximate lake locations/boundaries in both regions - Cook East (D) and Cook West (H). The model produced a dice similarity metric of 0.055 when comparing the overlap between true and predicted masks. While the model can find lakes in the center, it fails to find lakes elsewhere, i..e, Panel G has a lake in the SE corner of the panel that does not appear in the predicted lake mask.Therefore, modifications such as data augmentation, image fitting and cropping, and applying a dynamically lower learning rate were applied to improve model performance.


Mean Elevation Anomaly
Standard Deviation Anomaly
Wilson et al. (2025) Lake Mask
Predicted Lake Mask
Cook East
Cook West
(A)
(B)
(C)
(D)
(E)
(F)
(G)
(H)
Y (km)
X (km)
Elevation Anomaly (m)
Std Dev (m)